Heart-Disease Risk: Domain-Structured Bayesian Network vs Neural Network

Phase 1: Conventional Machine Learning and a Provisional Bayesian Network

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***Abstract*—This Phase 1 project studies historical heart-disease presence using a UCI-derived, four-hospital dataset. The supplied 920 records yield 918 after exact-record deduplication. We compare Logistic Regression, Random Forest, Histogram Gradient Boosting, XGBoost and a provisional discrete Bayesian Network under common data partitions and training-fold preprocessing. Five outer folds with three-fold inner tuning select Random Forest by mean validation F1 of 0.8358. A separate development partition selects its probability cutoff of 0.38. On 184 previously inspected retrospective evaluation records, this model achieves recall 0.9608, F1 0.8448 and ROC-AUC 0.9029. Logistic Regression achieves higher retrospective F1 and AUC, but evaluation results do not replace the development-selected family. Independent development-only hospital checks reveal weak transfer: the selected Swiss model trails an always-positive reference, while the VA Long Beach model has specificity 0.05. We discuss model strengths, threshold trade-offs, missingness and the limitations of a manually specified Bayesian graph. The work provides reproducible academic evidence rather than clinical validation or future-event prediction. Neural-network implementation belongs to Phase 2.**

***Index Terms*—heart disease, Bayesian network, Random Forest, nested cross-validation, missing data, retrospective evaluation**

I. Introduction and Research Objectives

The assigned project compares a domain-structured Bayesian Network (BN) with a neural network. This report covers the Machine Learning evaluation in Phase 1. It implements a probabilistic classifier and four conventional baselines. The neural-network comparison is reserved for Phase 2. The present outcome is recorded disease presence, so the word "risk" in the assigned title must not be interpreted as a validated estimate of a future cardiac event.

The UCI Heart Disease archive contains measurements from Cleveland, Hungary, Switzerland and VA Long Beach [1]. These historical cohorts differ in disease prevalence and measurement completeness. A model can obtain a high positive-class score in a cohort dominated by positive cases while identifying few negative cases. Consequently, a useful comparison needs explicit partition rules, multiple metrics and an honest account of population differences.

We investigate three research questions. First, which of five fixed model families performs best under a common development F1 criterion? Second, what changes when a development-selected probability threshold replaces the default 0.50 cutoff? Third, how does a procedure selected independently on other hospitals perform on the excluded hospital? We also examine what the BN's graph explains and which independence assumptions remain unsupported.

The contribution is a reproducible comparison with exact record memberships, nested search traces, saved probabilities and verified model files. We do not claim a new learning algorithm or universal superiority of one family. Model explanations below distinguish established algorithmic mechanisms from interpretations that this experiment has not tested through ablation.

II. Literature Review and Research Gap

Breiman's Random Forest construction combines randomized trees and relates generalization to tree strength and their correlation [2]. This motivates an ensemble baseline for nonlinear relationships. It does not establish that a forest will outperform a simpler classifier on this particular dataset. Logistic Regression provides a regularized linear log-odds baseline against which additional model complexity can be judged.

Chen and Guestrin describe XGBoost as a scalable regularized tree-boosting system [3]. Histogram Gradient Boosting provides another boosting implementation based on binned features [4]. These are complementary algorithm families rather than interchangeable replications. Their original large-scale motivations do not guarantee an advantage on a cohort of fewer than one thousand records.

Friedman, Geiger and Goldszmidt study BN classifiers and Tree Augmented Naive Bayes (TAN) [5]. Their work motivates explicit probabilistic factorization and relaxing restrictive conditional-independence assumptions. Our manual one-edge extension is not their learned TAN algorithm. A functioning BN and normalized conditional probability tables do not independently validate the medical meaning of its arrows.

Clinical context is narrower than graph validation. Fletcher et al. discuss exercise ST depression through both its magnitude and shape [6]. This motivates considering oldpeak and slope jointly as a modeling inference. Their discussion of age-dependent peak heart rate also challenges treating age and achieved heart rate as conditionally independent. This context does not validate our exact DAG, quantile bins or fitted CPTs; it identifies assumptions that deserve explicit review.

TABLE I
RETAINED COHORT BY HOSPITAL, COMPUTED FROM THE SUBMITTED CSV

Hospital	Records	Positive	Positive (%)
Cleveland	303	139	45.9
Hungary	293	106	36.2
Switzerland	123	115	93.5
VA Long Beach	199	148	74.4
Combined	918	508	55.3

Nested validation separates hyperparameter selection from the folds used to estimate configuration-selection performance [7]. Even with nested tuning, choosing the best family from several outer-CV scores introduces family-selection optimism. Calibration literature also distinguishes useful ranking from reliable absolute probabilities [8]. Accordingly, we report discrimination, threshold-dependent decisions and probability error separately.

The project's gap is a fair, traceable comparison on the supplied combined cohort, including the cost of high recall and the limits of transfer across hospitals. Published results on other targets, splits or cohorts are contextual evidence, not directly comparable numerical benchmarks for this study.

III. Dataset Quality and Exploratory Analysis

A. Source, target and record integrity

The supplied CSV contains 920 records and 16 columns. UCI supplies the underlying four hospital databases and variable definitions [1]; the local CSV uses renamed categories and the feature name `thalch` for achieved maximum heart rate. The exact acquisition and merge history of the supplied compilation remains incomplete. The dataset citation is retained, and unavailable patient linkage is disclosed rather than inferred from row identifiers.

The binary label is

$$Y = \mathbf{1}(\texttt{num} > 0), \qquad \texttt{num} \in \{0, 1, 2, 3, 4\}. \tag{1}$$

This mapping discards severity. Validation rejects invalid targets, missing or repeated row IDs and malformed required fields. Comparing all original columns except the arbitrary row ID removes two later exact-record duplicates. The retained 918 records contain 508 positive and 410 negative labels. Deduplication alone cannot prove that records represent independent patients.

Table I shows why combined class balance hides substantial site differences. The correct Swiss and VA positive counts are 115 and 148, respectively; these counts follow the dataset and structured audit rather than an older narrative table. The supplied source already includes the documented correction of ID 304's hospital label from Cleveland to Hungary. The current run uses this corrected grouping.

B. Missingness and eligible predictors

We retain five numeric predictors: age, resting blood pressure, cholesterol, achieved maximum heart rate and ST depression. Six categorical predictors describe sex, chest-pain type, fasting blood sugar, resting ECG, exercise angina and ST slope. ID, hospital and target cannot enter the feature matrix. Hospital is available exclusively for partitioning and descriptive analysis. The features `ca` and `thal` are excluded because 609 and 484 of the retained records are missing them, with strong differences across hospitals.

The deterministic cleaning rule treats zero blood pressure and cholesterol as missing entries. After cleaning, missing counts are pressure 60, cholesterol 201, fasting sugar 90, resting ECG 2, achieved heart rate 55, exercise angina 55, ST depression 62 and ST slope 307. Age, sex and chest-pain type are complete. Missingness may reflect site-dependent measurement practice, so imputation cannot be assumed to remove the associated bias.

EDA includes target counts, site prevalence, numeric distributions, categorical breakdowns, missingness and correlations. These describe the observed cohort. Correlation does not establish causal effects or measure an individual predictor's contribution to a fitted ensemble. Exercise-test variables also constrain the measurement stage at which the inputs could be available.

IV. Feature Engineering and Model Formulation

A. Training-fold transformations

Each conventional model receives the same eleven raw predictors. A numeric pipeline learns median imputations, adds missing-value indicators and standardizes the resulting values. A categorical pipeline learns the most frequent values and then applies one-hot encoding, tolerating valid categories absent from a training fold. Only the permitted training rows fit these transformations. Validation, cutoff and evaluation rows use the already fitted transformations.

Scaling supports the regularized linear baseline. The tree families generally do not need scaling for split ordering, but the shared pipeline keeps the preprocessing contract consistent. Although HGB can handle missing values natively, this experiment retains explicit common preprocessing. We have not measured whether a different family-specific preprocessing choice would improve its performance.

B. Conventional model families

For Logistic Regression, the modeled class probability is

$$P(Y = 1 \mid \mathbf{x}) = \frac{1}{1 + \exp[-(\beta_0 + \boldsymbol{\beta}^T \mathbf{x})]}. \tag{2}$$

Regularization controls coefficient complexity. The final selection-partition search chooses $C = 0.2$ and an L1 penalty. Coefficients offer a compact explanation of the fitted log-odds relation, but correlated predictors and the encoding must be considered when interpreting them.

Random Forest averages class probabilities from randomized trees. Bootstrap sampling and random feature subsets diversify trees; averaging can reduce variance. The final search chooses 200 trees, maximum depth 5, minimum leaf size 2 and `sqrt` feature sampling. Limited depth constrains individual-tree complexity. These mechanisms plausibly suit nonlinear

tabular relationships, but their causal contribution to this run's score was not established by an ablation.

HGB and XGBoost add trees sequentially to improve a loss objective. The chosen HGB configuration has learning rate 0.03, seven leaf nodes and L2 regularization 1. XGBoost uses 150 trees, depth 2 and learning rate 0.03. These settings come from training-only search. A low learning rate and restricted tree size control the candidate's complexity; they are not proof of optimality beyond the declared grid.

C. Provisional Bayesian Network

The BN contains disease Y as parent of each feature and an additional edge from oldpeak O to slope L. Its factorization is

$$P(Y, \mathbf{X}) = P(Y)P(L \mid Y, O) \prod_{i \neq L} P(X_i \mid Y). \quad (3)$$

The extra edge allows one pair of related ST descriptors to depart from class-conditional independence. All other omitted dependencies remain explicit simplifying assumptions. Arrow direction specifies a statistical factorization, not a causal claim that disease produces age or recorded sex. The graph remains provisional and its adequacy for the assigned domain-structure requirement needs review.

Within each training partition, the BN learns imputations and one-third/two-thirds numeric quantile bins. Fixed categorical vocabularies preserve known states. Training-only counts estimate smoothed conditional probability tables (CPTs):

$$\widehat{P}(X_i = k \mid \mathrm{Pa}_i = j) = \frac{N_{ijk} + \alpha}{N_{ij} + K_i \alpha}, \quad \alpha = 1. \quad (4)$$

Here K_i is the number of child states. Pseudocounts avoid zero estimates for unseen configurations. Inference marginalizes missing query evidence and normalizes class probabilities. For example, observing slope without oldpeak sums $P(O \mid Y)P(L \mid Y, O)$ over oldpeak states. The evaluated pooled graph has 49 free parameters. Binning loses numeric resolution, and imputation before counting does not model the full uncertainty in missing training data.

V. Experimental Design and Validation

A. Partitioning and nested model selection

The historical stratified 80/20 split, seed 42, gives development $D = 734$ and retrospective evaluation $T = 184$. The source project's T rows had already been inspected. This run preserves their identity and reports retrospective performance; it cannot create a new untouched test through resplitting.

A stratified 75/25 split of D, seed 43, gives selection $S = 550$ and cutoff $C = 184$. Five outer stratified folds, seed 44, assess each family on S. Inside every outer training fold, three inner folds, seed 45, select a configuration by mean F1 at cutoff 0.50. Mean AUC and canonical parameter ordering break exact inner ties. The outer score is the unweighted mean of its five validation-fold scores.

The family rule chooses highest outer mean F1, then AUC, followed by the fixed order BN, LR, RF, HGB and XGB if exact ties remain. A final three-fold search, seed 46, fits each family on S only. There is no S+C refit. Search budgets are LR 8, RF 6, HGB 8, XGB 8 and BN 1 configurations. BN's single alpha setting and unequal grids limit what can be concluded about relative tuning opportunities.

B. Thresholds, metrics and uncertainty

Each frozen S-fitted model predicts C. A grid of 101 cutoffs from 0.00 through 1.00 maximizes C's F1. Ties choose the cutoff closest to 0.50, then the lower value. Decisions use $p \geq t$. Family selection remains fixed before C or T scores are available.

For positives TP, missed positives FN, false positives FP and true negatives TN,

$$F_1 = \frac{2TP}{2TP + FP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}. \quad (5)$$

Accuracy, precision, recall, ROC-AUC, average precision, Brier score, log loss and confusion matrices complement F1. ROC-AUC assesses ranking across cutoffs, while recall and specificity describe a particular operating point. Brier score measures squared probability error and alone does not establish calibration. Saved reliability bins provide descriptive calibration evidence.

Paired row bootstrap uses 2,000 resamples and percentile 95% intervals on saved evaluation predictions. These intervals condition on the fitted models and split. They omit training-selection variability and any unknown repeated-patient clustering, and do not constitute a formal model-superiority test.

C. Independent hospital procedures

Each hospital experiment uses D only. It excludes that hospital's D records, splits the other hospitals into local selection/cutoff sets and repeats the entire family-selection, fitting and threshold procedure. The held hospital and T influence no local decisions. No pooled fitted model, preprocessing state, winner or cutoff transfers. Within the remaining hospitals, CV still stratifies rows rather than making every fold hospital-disjoint. These experiments test transfer across supplied groups, not prospective independent clinical validation.

VI. Results and Model Interpretation

A. Which model was best, and under which criterion?

Random Forest is the selected model because its outer mean F1 of 0.8358 is the highest under the predefined development rule. XGBoost follows at 0.8308, a difference of approximately 0.0049. That small difference does not establish statistically significant or universal superiority. The winning CV score also retains optimism from selecting across families.

Logistic Regression is the strongest retrospective scorer on accuracy, precision, F1, AUC and Brier error in Table II. Its F1 is 0.8531 and AUC is 0.9125. These results make it a strong simpler comparator, but using T to replace RF would violate the selection protocol. "Best" therefore has a precise meaning: RF wins the development F1 rule, while LR leads several descriptive retrospective metrics. BN has the highest mean outer-CV AUC (0.8780), further illustrating that a different objective could yield a different ranking.

TABLE II
FIVE-FAMILY COMPARISON. NESTED CV USES S AT CUTOFF 0.50. REMAINING COLUMNS USE EACH C-SELECTED CUTOFF ON T (184 PREVIOUSLY INSPECTED RECORDS: 102 POSITIVE, 82 NEGATIVE).

Model	CV F1	Cutoff	Accuracy	Precision	Recall	Specificity	F1	ROC-AUC	Brier
Logistic Regression	0.8231	0.51	0.8315	0.8257	0.8824	0.7683	0.8531	0.9125	0.1227
Random Forest	**0.8358**	0.38	0.8043	0.7538	0.9608	0.6098	0.8448	0.9029	0.1308
Histogram Gradient Boosting	0.8104	0.37	0.8152	0.7698	0.9510	0.6463	0.8509	0.8945	0.1318
XGBoost	0.8308	0.34	0.8043	0.7463	0.9804	0.5854	0.8475	0.9039	0.1273
Bayesian Network	0.8179	0.54	0.7880	0.8058	0.8137	0.7561	0.8098	0.8765	0.1544

B. How Random Forest performed well

At cutoff 0.38, RF correctly identifies 98 of 102 positives and 50 of 82 negatives. Its confusion matrix is $[[50, 32], [4, 98]]$, ordered as $[[TN, FP], [FN, TP]]$. Recall is 96.08%, precision is 75.38%, F1 is 84.48% and specificity is 60.98%. AUC of 0.9029 indicates useful retrospective ranking in this cohort, rather than medical validity of the threshold.

The forest can represent feature interactions without assuming linear log-odds. Averaging randomized, depth-limited trees provides a plausible explanation for its competitive development F1. We did not run an ablation proving that nonlinear interactions, depth restriction or a particular feature caused the observed advantage. No feature-importance ranking is claimed.

RF's conditional 95% bootstrap intervals are F1 [0.7895, 0.8915], AUC [0.8571, 0.9432] and recall [0.9192, 0.9910]. These relatively broad ranges support restrained interpretation. They do not resolve cohort selection or generalization uncertainty.

C. Threshold and model trade-offs

The lower RF cutoff prioritizes detecting positive labels while accepting additional false-positive decisions. At cutoff 0.50, the same forest misses 13 positives and makes 23 false positives; at 0.38, it misses four and makes 32. Thus nine fewer missed positives accompany nine additional false alarms on these rows. This is a statistical F1 operating point, not a medically established threshold. XGBoost detects 100 of 102 positives but makes 34 false positives, whereas LR detects 90 positives and makes 19 false positives. RF sits between these two error profiles with 98 positives detected and 32 false positives. None of these counts alone determines the preferred clinical operating point.

LR offers a smaller, easier-to-inspect parameterization and strong probability error in this run, but its linear decision surface can miss interactions unless explicitly represented. RF captures nonlinear patterns at the cost of a larger ensemble and less direct coefficient interpretation. Boosting offers controlled sequential correction but introduces additional tuning choices. BN offers an explicit factorization and exact missing-evidence queries, while its independence assumptions, discretization and unvalidated structure constrain accuracy and interpretation. Algorithm complexity and search time are distinct from demonstrated usefulness.

TABLE III
SELECTED-MODEL HOSPITAL CHECKS, EACH FITTED INDEPENDENTLY USING D ONLY

Hospital	N	Model	F1	Ref. F1	Spec.
Cleveland	248	LR	0.7951	0.6260	0.7481
Hungary	229	RF	0.6948	0.5367	0.6207
Switzerland	96	RF	0.8712	0.9565	0.5000
VA Long Beach	161	LR	0.8561	0.8582	0.0500

Ref. F1 uses always-positive decisions on the same evaluation rows. Spec. denotes specificity.

D. Hospital shift and failure cases

Table III reports the local development-selected family for each hospital, rather than selecting a family from that hospital's evaluation ranking. Switzerland contains 88 positives and only eight negatives in its 96 evaluation rows. RF obtains F1 0.8712, below the always-positive reference of 0.9565. VA Long Beach's LR has F1 0.8561 versus reference 0.8582 and rejects only two of 40 negatives. Its specificity is 0.05 and balanced accuracy is 0.5167, despite AUC 0.7079.

These are substantial limitations rather than results to hide. For positive prevalence p, an always-positive classifier has $F_1 = 2p/(1+p)$. This explains why a large F1 can coexist with poor negative-class performance. Evaluating only F1 would overstate the practical value of the Swiss and VA results. The observed failures do not authorize retuning on those already inspected hospital outcomes.

VII. CHALLENGES, LIMITATIONS AND REPRODUCIBILITY

A. Challenges and responses

Incomplete and heterogeneous measurements: Zero-coded and absent values require deterministic cleaning followed by fold-local imputation. Missing indicators retain some information about absent numeric values, but do not correct selection bias or establish a missing-at-random mechanism.

Data leakage during tuning: Learning preprocessing globally or choosing hyperparameters on validation rows would inflate estimates. We place transformations inside each fit and separate selection, cutoff and retrospective roles. Isolation tests trace fitting/prediction calls and check exact record membership.

Interpreting a provisional graph: Related ST measurements motivate considering an extra dependency, but not every omitted edge is justified. The register distinguishes assumptions from domain evidence. In particular, independence between

age and achieved exercise heart rate remains questionable. A revised graph would require new development evaluation and revised inference, not silent replacement of the evaluated model.

High recall versus false alarms: Lower cutoffs increase positive decisions. The pooled RF misses four positives but flags 32 negatives. Both errors must appear in the discussion. F1 optimization does not encode real clinical costs, which were not measured here.

Historical exposure and small samples: T had already been inspected, individual hospital groups have limited class support, and unknown patient linkage prevents a patient-level independence claim. Saved sex/age summaries and calibration bins remain descriptive. No prospective cohort, causal analysis or clinical-utility evaluation was performed.

B. Code, environment and evidence

The shared GitHub repository is AML Capstone Project. This report follows commit `16f1e33` and the recorded run 20261025T151903Z_ea0f799c, dated 25 October 2026. Source code separates data loading, baseline APIs, advanced preprocessing, nested training, BN inference, experiment persistence, verification and result summarization.

The completed run records 2,940 fits, five experiment groups and approximately 196 seconds of experiment execution on its recorded machine, excluding setup and later verification. That runtime is environment-specific and is not a comparative model-speed benchmark. Recorded versions include Python 3.14.2, NumPy 2.5.4, pandas 3.1.0, scikit-learn 1.9.1 and XGBoost 3.4.1. The run records its macOS OpenMP library path and dependency freeze.

An independent recheck passed 48 tests and 39 subtests. Verification reloads 25 saved models, checks their hashes and reproduces probabilities and metrics within 10^{-12}. It also reconstructs cutoff choices, fold scores and selection decisions, and checks partition/hospital isolation. These checks establish consistency of the submitted artifacts; they cannot validate clinical assumptions. Running tests explicitly from `tests/` avoids collecting duplicate historical test snapshots.

All five members appear in the supplied team roster. Commit history and saved experiment evidence support traceability, but this report does not assign unverified individual responsibilities. Each member must be able to explain the methods, implementation and limitations in the individual viva.

VIII. Conclusion

Phase 1 delivers a five-family historical disease-presence comparison with reproducible preprocessing, nested selection and separately selected thresholds. RF wins the predefined development F1 criterion. At its cutoff of 0.38 it identifies 98 of 102 positive retrospective records, with 32 false positives. LR leads several retrospective metrics, so RF must not be described as the best model under every objective.

Hospital checks expose weak transfer and demonstrate why specificity and a same-row constant reference are necessary beside F1. The BN provides genuine probabilistic estimation and inference, while its manually specified domain structure remains provisional. Phase 2 can add the assigned neural-network comparison under an explicit validation design. The current evidence supports an academic ML study, with no claim of clinical deployment readiness or future-event prediction.

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